

Linear Equations with Distribution and Fractions

Skills practised: Distributing first, clearing fractions with the LCD, variables on both sides, and equivalent rewrites.

Worked pattern — follow this order every time.

Solve $\frac{x}{2} + \frac{1}{3} = 2(x - 1)$.

Step 1 — clear fractions. The denominators are 2 and 3, so the LCD is 6. Multiply *every term on both sides* by 6: $3x + 2 = 12(x - 1)$.

Step 2 — distribute. Multiply through the brackets: $3x + 2 = 12x - 12$.

Step 3 — collect. Variable terms to one side, numbers to the other: $14 = 9x$.

Step 4 — isolate and check. $x = \frac{14}{9}$. Substitute back into the *original* equation to check.

Show every step. Leave answers as exact fractions where they are not whole numbers.

1. Solve for x : $3(x + 4) = 21$

Answer: _____

2. Solve for x : $-2(x - 5) = 14$

Answer: _____

3. Solve for x : $\frac{x}{3} + \frac{x}{6} = 5$

(Start by multiplying every term by the LCD.)

Answer: _____

4. Solve for x : $4 - 2(x + 3) = 10$

(The subtraction sign in front of the brackets belongs to the 2.)

Answer: _____

5. The expression $\frac{x}{4} + \frac{2}{3}$ appears on one side of an equation. Multiply it by the LCD, 12, and write the equivalent expression with no fractions left in it.

Answer: _____

6. Solve for x : $3(x - 2) = 2(x + 4)$

Answer: _____

7. Solve for x : $\frac{3x}{4} - 2 = \frac{x}{2}$

Answer: _____

8. Solve for x : $\frac{1}{2}(6x - 8) = 11$

Answer: _____

9. Expand and simplify $7 - 3(2x - 5)$, writing your answer in the form $a + bx$ or $bx + a$.

Answer: _____

10. Solve for x : $\frac{2}{3}(x + 6) = \frac{1}{2}(x + 2) + 5$

Answer: _____

11. Solve for x : $\frac{x + 3}{2} - \frac{x - 5}{4} = 5$

(A fraction bar groups the whole numerator — keep the brackets when you multiply.)

Answer: _____

12. Challenge. Solve for x , then check your solution in the original equation: $\frac{2}{5}(10x - 15) = 3(x - 1) + 4$

Answer: _____